Titration –I

(Link $KMnO_4$)Vs (Standard FAS solution)

s.no	Volume of standard FAS (ml)	Burette readings		Concordant value (Volume of $KMnO_4$) (ml)
		Initial (ml)	Final (ml)	
1	20			
2	20			
3	20			

Calculation :

Volume of $KMnO_4$ (link) solution V_1 = ml

Normality $KMnO_4$ (link) solution N_1 = ? N

Volume of standard FAS solution V_2 = 20 ml

Normality of standard FAS solution N_2 = 0.1 N

According to normality equation:

$$V_1 \times N_1 = V_2 \times N_2$$

$$N_1 = \frac{V_2 \times N_2}{V_1} =$$

Normality $KMnO_4$ (link) solution N_1 = _____ N

Titration–II

(Unknown oxalic acid) Vs (Link $KMnO_4$)

Burette is washed with water, rinsed with $KMnO_4$ solution and filled with same $KMnO_4$ solution up to the zero mark. Exactly 20 ml of unknown oxalic acid solution is pipetted out into the clean, washed conical flask. To this oxalic acid solution approximately 20ml of 2N sulphuric acid is added. This mixture is heated to 60 – 70^0C using Bunsen burner and that hot solution is titrated against $KMnO_4$ Link solution from the burette. $KMnO_4$ is added drop wise till the appearance of permanent pale pink colour. Burette reading are noted, the same procedure is repeated to get concordant values.

organic analysis.indd 272 19-01-2021 12:24:24